

any way for GEM to match the varying risk profiles of different investors.

To answer that, we use the Markowitz-Tobin fund separation theorem, which separates the decision of what assets to hold from the decision of how much risk to assume.¹² More specifically, this says that investors should hold the singular portfolio of assets having the highest Sharpe ratio, but those wanting a lower-risk profile should combine that optimal portfolio with a risk-free (or low-risk) alternative asset. Conversely, aggressive investors wanting higher returns can borrow in order to leverage the optimal portfolio.

Figure 8.15 shows the separation theorems tangency portfolio, where a straight line (the capital market line) originating from the risk-free rate just touches the efficient frontier of risky portfolios. Risk-averse investors will have a better reward-to-risk ratio and will be better off investing along this line rather than on the efficient frontier itself.

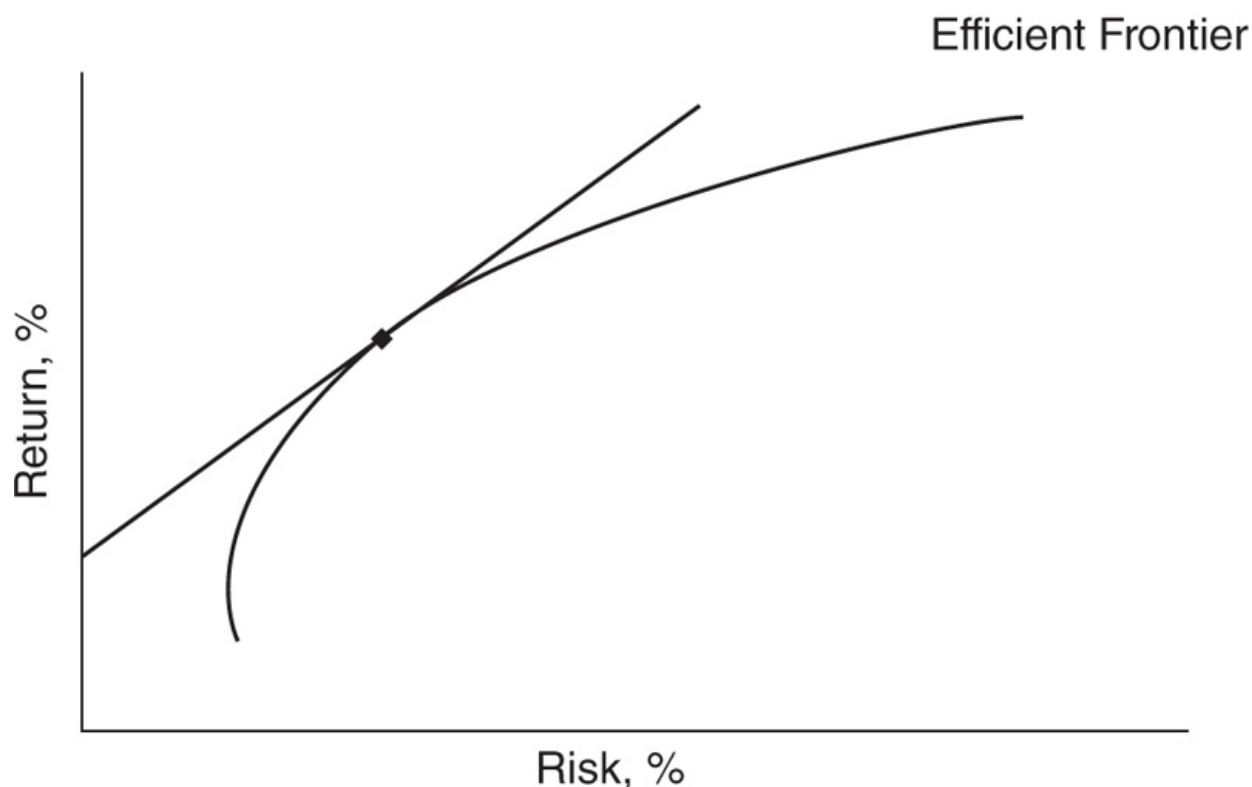


Figure 8.15 Capital Market Line and the Efficient Portfolio Set

Table 8.10 and Figure 8.16 show GEM leveraged 30% to suit aggressive investors who choose to borrow at the federal funds rate plus 25 basis points.